Foresight: Leakage-Controlled Vision Transformer for Visual Produce-Spoilage Estimation with Human-Validated Decay Labels

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***Abstract*—Image-based spoilage classifiers are often evaluated on random splits of datasets that contain duplicate images and repeated views of the same specimen, which inflates reported accuracy. We present Foresight, a single-photo produce-spoilage estimator built on the public TriModal Ripeness 6 (TR-6) dataset. An audit showed that all 10,796 *Classified* images are byte-identical copies of *Normal* images, and that 170 contents carry contradictory labels. After de-duplication, we split 5,335 eligible sRGB images 60/20/20 by capture session. A fine-tuned ViT-B/16 reaches a test spoilage macro-F1 of 0.954 (95% session-bootstrap CI 0.903–0.988), ROC-AUC 0.989, and 100% produce-type accuracy. Controlled three-seed ablations show three effects:**

- **conventional augmentation gives a small but consistent gain (+0.013);**
- **class-conditional GAN augmentation gives no reliable gain;**
- **evaluating on unseen capture days lowers macro-F1 by 0.098.**

A human review of 165 images found visible decay in only 72 of 109 images labelled *Spoiled*. This shows that the dataset label partly encodes storage time rather than visible decay. The model is deployed in a tested FastAPI/React application with three further outputs, each labelled in the app as what it is: attention-rollout explanations, a visible-decay check trained on human labels, and pilot segmentation and proxy shelf-life outputs.

***Index Terms*—Food spoilage, vision transformer, data leakage, explainable AI, GAN augmentation, dataset auditing.**

I. Introduction

Post-harvest loss of fruit and vegetables is a major contributor to food waste. Visual inspection is the most common triage method but is subjective. Deep image classifiers can automate this triage. However, their reported accuracies depend heavily on how training and test images are separated. When a dataset contains repeated views of the same specimen, a random split places near-identical images in both partitions, and the test score then measures memorisation rather than generalisation.

This paper studies spoilage estimation on TR-6 [1] with an emphasis on evaluation validity. Our contributions are:

1) an audit of TR-6 that quantifies duplicate and same-session leakage under different split strategies;
2) a fine-tuned vision transformer [2] evaluated on a duplicate-aware, session-grouped split, with bootstrap confidence intervals;
3) controlled multi-seed ablations of augmentation, GAN augmentation [6], [7], split design, noise robustness and denoising [8];
4) a human validation of the dataset's spoilage labels;
5) an integrated, tested application that labels pilot and proxy outputs as such.

TABLE I
Test-to-train leakage under different split strategies

Split strategy	Exact dup.	Same session	Same day
Random, raw files	59.8%	100.0%	100.0%
Random, de-duplicated	0.0%	99.9%	100.0%
Session-grouped (ours)	0.0%	0.0%	76.2%
Day-grouped (sensitivity)	0.0%	0.0%	1.5%

II. Dataset and Leakage Control

TR-6 contains 21,653 files (19.8 GiB): 10,676 sRGB images, 10,950 thermal IR-fusion images and 18 methane logs. They cover six produce types: banana, carrot, guava, Indian gooseberry, mango and tomato. Each fruit was photographed from 17 viewpoints per capture session over consecutive days.

Audit. Hashing every file with SHA-256 yields only 10,776 unique image contents, because the *Classified* tree is a relabelled copy of the *Normal* tree. 170 contents (all thermal) appear under both labels. Within every produce type, all *Not spoiled* captures precede all *Spoiled* captures.

Eligibility and split.

- Conflicting contents were excluded rather than tie-broken. *Normal*-only contents were treated as unlabelled.
- Only sRGB was used, because the target application receives phone photographs. This left 5,335 eligible images.
- A capture session was defined as images of the same produce and label taken at most 300 s apart. Sessions were assigned 60/20/20 within each produce $\times$ label stratum (seed 42), giving 3,177/1,087/1,071 images and 189/64/64 sessions.

Table I shows why this matters. A random split of the raw files places an exact copy of 59.8% of test images in training, and a same-session view of every test image. The session split removes both kinds of leakage, but 76.2% of test images still share a produce and capture day with training. A day-grouped variant reduces that to 1.5% and is used as a sensitivity test.

III. Method

Model. We fine-tune a ViT-B/16 pretrained on ImageNet-21k and ImageNet-1k (86 M parameters, `timm` [10]). Two linear heads act on the CLS embedding z:

$$p_{\text{type}} = \text{softmax}(W_t z + b_t), \qquad p_{\text{spoil}} = \text{softmax}(W_s z + b_s). \tag{1}$$

The loss is $\mathcal{L} = \text{CE}(y_t, p_{\text{type}}) + \text{CE}(y_s, p_{\text{spoil}})$. Self-attention follows $\text{softmax}(QK^\top/\sqrt{d_k})V$.

Training.

- Optimiser: AdamW with weight decay 0.01, a backbone learning rate of 2×10^{-5} and a head learning rate of 2×10^{-4}, one warm-up epoch and cosine decay.
- Schedule: batch 16, 12 epochs, 224×224 input.
- Augmentation: crop, flips, $\pm 15°$ rotation and ± 0.15 brightness/contrast. Hue and saturation are left unchanged so that colour cues of decay are preserved.
- Selection: the checkpoint is chosen by validation composite macro-F1 (epoch 9 of 12), and the test set is scored once.
- Hardware: training took 54.7 min on an Apple M5 GPU (MPS).

Explanations. Attention rollout [3] multiplies head-averaged, residual-corrected attention matrices across the 12 blocks.

Additional outputs. Three lightweight models reuse the frozen embedding z or the image:

1) a logistic *visible-decay* probe trained on human labels (Section V);
2) a ridge regressor for a *proxy* shelf-life, defined as the days until the series' first *Spoiled* capture;
3) a pilot U-Net [4] with a ResNet-18 encoder [5] for fruit and decay masks, trained with $0.5\,\text{BCE} + 0.5\,\text{Dice}$ on machine-generated masks.

IV. Results

Main result. On the 1,071 test images the deployed model achieves:

- spoilage accuracy 0.961, macro-F1 0.954 and ROC-AUC 0.989;
- a session-level bootstrap 95% CI for macro-F1 of 0.903–0.988;
- 100% produce-type accuracy (Fig. 1).

The *Spoiled* class has precision 0.981 but recall 0.894: 36 of 340 test images are missed, 27 of them Indian gooseberries. The spoilage head has an ECE of 0.038, yet 88% of its errors are made with confidence of at least 0.9.

Baselines and ablations (Table II). Colour-histogram and frozen-ViT linear probes reach macro-F1 0.828 and 0.916. All ViT-S/16 conditions use identical settings and three seeds.

- Conventional augmentation improves macro-F1 by +0.013, positive on all seeds.
- Adding 480 GAN images changes it by -0.009, with an inconsistent sign across seeds.

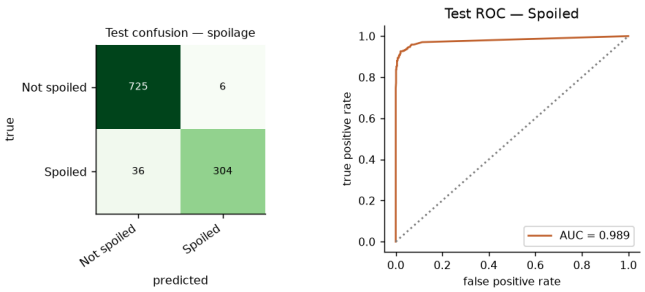


Fig. 1. Test confusion matrix and ROC curve for the spoilage head.

TABLE II
Test spoilage macro-F1 (ViT-S: mean ± SD of 3 seeds)

Model	Macro-F1	ROC-AUC
Colour histogram + LogReg	0.828	0.885
Frozen ViT-B/16 + LogReg	0.916	0.979
ViT-S/16, no augmentation	0.914 ± 0.006	0.971
ViT-S/16, conventional aug.	0.928 ± 0.007	0.981
ViT-S/16, conv. aug. + GAN	0.919 ± 0.018	0.976
ViT-S/16, conv. aug., day split	0.830 ± 0.007	0.909
ViT-B/16 (deployed)	**0.954**	**0.989**

- The GAN images are of limited quality: the real-data classifier recognises the intended produce in only 34.6% of them. No memorisation was found; the median nearest-neighbour distance is 22.5, against 12.8 between real images.
- The same configuration evaluated on the day-grouped split loses 0.098 macro-F1 (0.830 ± 0.007), which shows that session-level results are optimistic for unseen days.

Noise robustness (Fig. 2). With deterministic test corruptions, spoilage macro-F1 drops:

- to 0.805 under Gaussian noise ($\sigma = 0.2$);
- to 0.776 under salt-and-pepper noise ($p = 0.1$);
- to 0.800 under speckle noise ($\sigma = 0.5$).

Produce accuracy stays at or above 0.96 in every condition. A 3×3 median filter restores salt-and-pepper performance to 0.934. A resumed DnCNN gives small gains only under heavy noise. The intrinsic noise of TR-6 ($\sigma \approx 0.003$) is far below these stress levels.

Explainability. Rollout maps concentrate on the fruit (Fig. 3).

- *Faithfulness:* masking the top-20% rollout patches lowers predicted-class confidence by 0.192, against 0.020 for random patches (Wilcoxon $p = 2.3 \times 10^{-30}$, $n = 240$).
- *Stability:* maps are stable under horizontal flip (Spearman $\rho = 0.95$) and mild noise ($\rho = 0.96$), against 0.59 between different images.
- *Artefacts:* isolated bright background tokens match known ViT register artefacts [9].
- *K-means:* with $k = 6$ on frozen ViT embeddings, clusters recover produce type (NMI 0.95) but not spoilage (NMI 0.14).

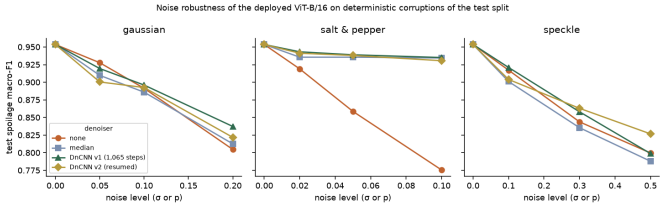


Fig. 2. Spoilage macro-F1 under synthetic noise with and without denoising.

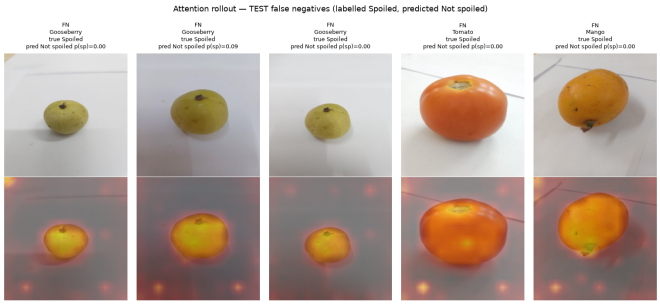


Fig. 3. Attention rollout on test false negatives: *Spoiled*-labelled fruit without visible decay.

V. Human Validation of Spoilage Labels

A stratified set of 180 images was reviewed in Label Studio, giving 165 definite decisions.

- Visible decay was seen in 72 of 109 *Spoiled*-labelled images and in 6 of 56 *Not spoiled* ones, an agreement of 73.9%.
- Agreement is high for banana and carrot (20/20 each) but low for tomato (2/20).
- This explains the model's tomato and gooseberry false negatives: the folder label marks storage time, which can precede visible decay.

A visible-decay probe trained on 102 of these decisions reaches 0.931 test accuracy (95% CI 0.78–0.98, $n = 29$; AUC 0.986). Using the spoilage head directly as a decay detector reaches 0.897 accuracy (AUC 0.881).

VI. System and Additional Outputs

Application. The React UI posts a photo to a FastAPI service, which validates the upload, runs all models and stores the result in SQLite without keeping the image.

- Median end-to-end latency is 48 ms on the M5 GPU; a ViT-B forward pass takes 201 ms on CPU.
- 37 backend/ML tests and 7 frontend tests pass, and a live end-to-end test passes 11/11 checks.
- Of 60 held-out full-resolution images, the live API classified produce type correctly for 60 and spoilage for 57.

Additional outputs.

- The proxy shelf-life regressor reaches a test MAE of 4.29 days ($R^2 = 0.89$), against 9.95 days for a per-produce mean.
- The pilot U-Net agrees with its machine-generated reference masks with fruit Dice 0.98 and decay Dice 0.59 ($n = 63$).

Both outputs are labelled as proxy or pilot in the interface. They are not validated against measured shelf life or human-drawn decay masks.

VII. Limitations

- **Specimens:** TR-6 lacks specimen identifiers, so the same physical fruits appear in training and test.
- **Domain:** all images are lab photographs on plain backgrounds; real-world photos were not evaluated.
- **Labels:** the spoilage labels are date-based.
- **Annotation:** the human labels come from a single annotator.
- **Seeds:** the deployed model was trained with one seed.

VIII. Conclusion

On TR-6, evaluation design matters more than model choice: duplicate-aware, session-grouped splitting is essential, and day-level generalisation remains notably harder. A fine-tuned ViT-B/16 gives strong, explainable spoilage estimates in a working application. Human validation shows that the dataset's *Spoiled* label must be interpreted as a storage-stage label. Future work: human-drawn decay masks for a validated U-Net, real-world test photos, confidence calibration and public deployment.

References

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